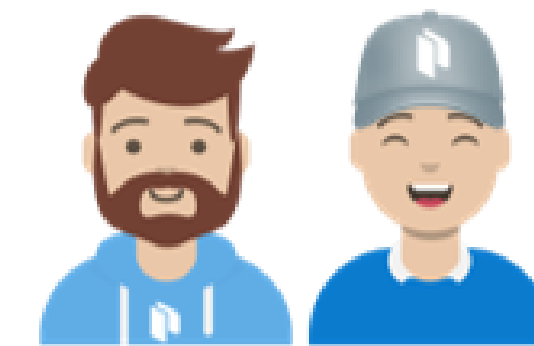
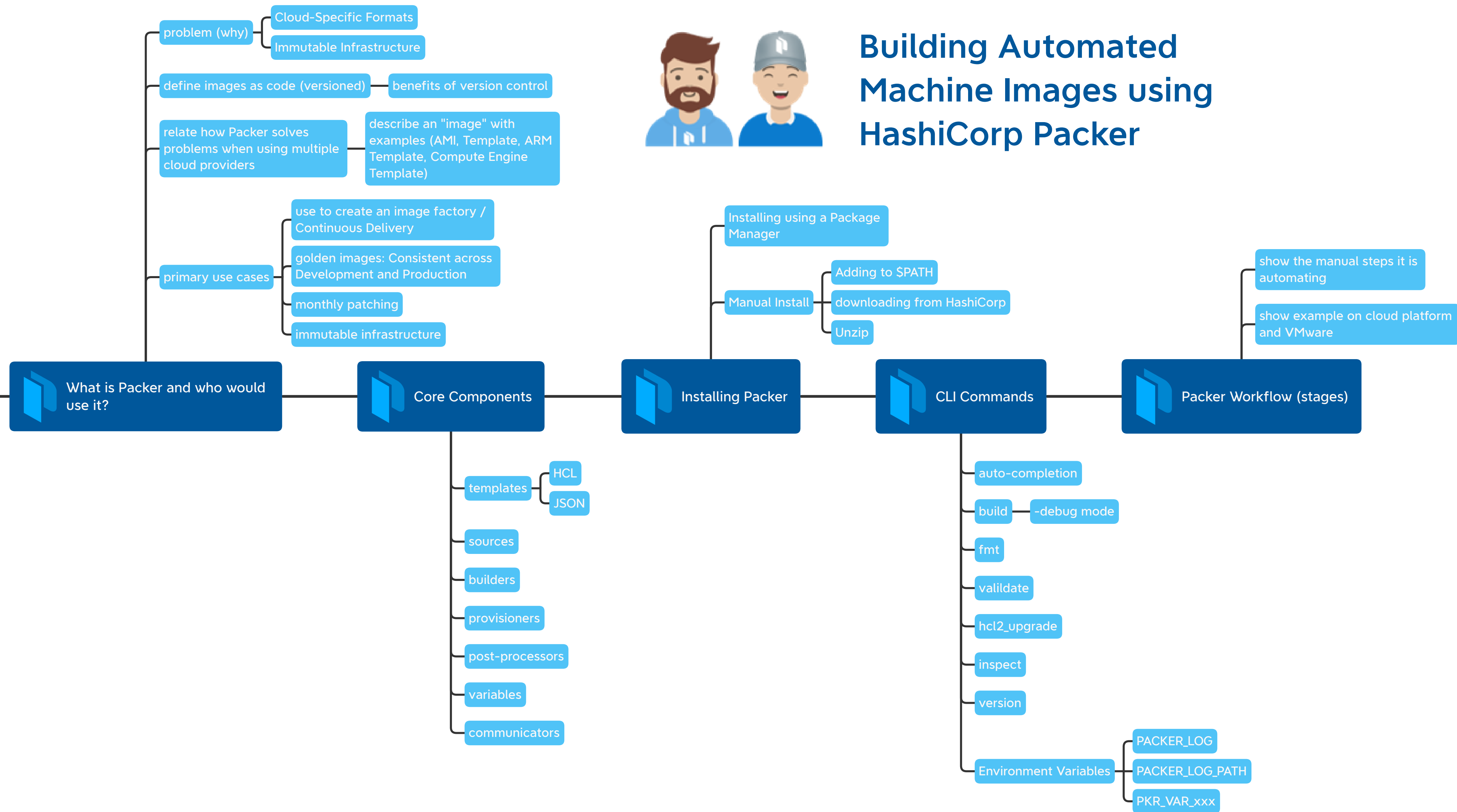


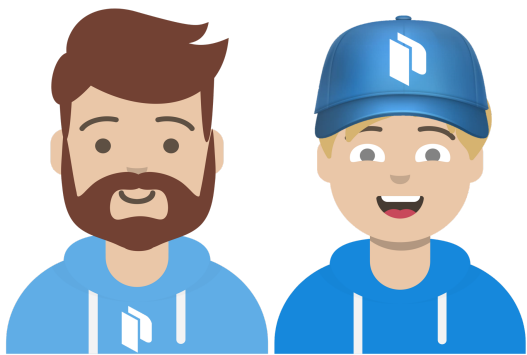


Building Automated Machine Images using HashiCorp Packer

Section 1: Introduction to HashiCorp Packer



Building Automated Machine Images using HashiCorp Packer



Section 2: Writing Packer Templates

Configuration Syntax

HCL2

- suggested format for templates moving forward in Packer 1.7+
- HCL is designed to strike a balance between human-readable and machine-parsable
- Terraform User? This looks familiar!!!
- Configuration format is VCS friendly (multi-line lists, trailing commas, and auto-formatting)

file naming

.pkr.hcl

Any files with a different extension are ignored

syntax constructs

blocks

arguments

JSON

old way of writing templates

HCL2 vs. JSON

HCL Syntax

block organization

Commenting

/ xxx /

#

//

syntax highlighting

Plugins for Terraform/HCL exist for most major editors,

interpolation syntax

Declarative

references to other resources do not depend on the order they are defined

Configuration can be in a single file or split across multiple files.

Plugins

allow new functionality to be added to Packer without modifying the core source code.

Working with Packer's Core Components in HCL

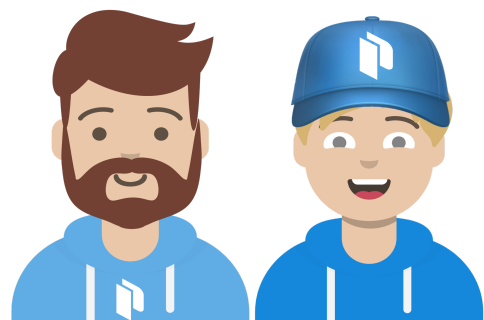
Source Blocks

Builders

Variables

Post-Processors

Provisioners



Building Automated Machine Images using HashiCorp Packer



Section 3 - Builders



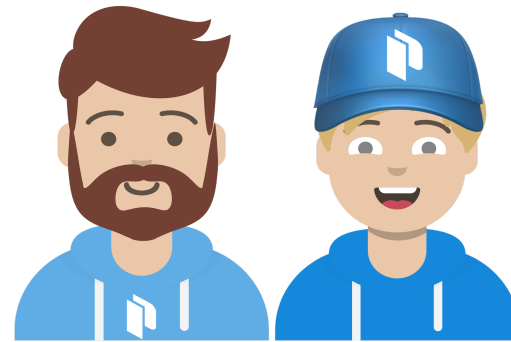
What Are Builders?

- responsible for building images
- can use one or more in a Packer template
- generally refers to a source block
- Popular builders
 - aws
 - azure
 - vmware
 - docker
 - google cloud
 - null
 - vagrant
- HCL examples



Use Cases and Examples

- AWS
 - EBS-Backed
- Azure
 - ARM
- GCP
- VMware
 - iso
 - clone



Building Automated Machine Images using HashiCorp Packer

Section 4: Variables

Syntax

- Works a lot like variables from other programming languages
- Allows removal of hard coded values
- Makes configuration easier to understand and re-use

Types of Variables

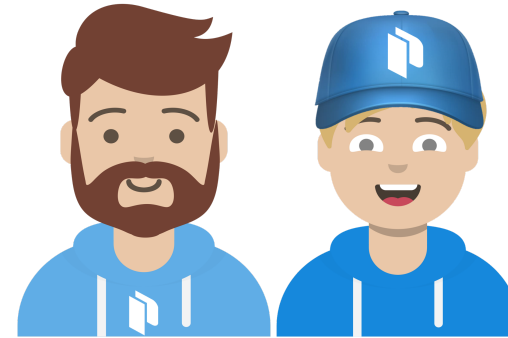
- String
- Number
- List
- Map
- Boolean

Declaring Variables

- using a .pkrvars.hcl file
- declare on the CLI
- inside the main template itself
- declaring the type
- validation

Using Variables in Templates

- Default Values
- Environment Variables
- Definition files
- CLI Prompts
- Locals
- Order of Operations



Building Automated Machine Images using HashiCorp Packer

Section 5: Provisioners

Intro to Provisioners

- install packages
- run scripts
- upload files
- shell commands

3rd Party Integrations

- Ansible
- Chef
- Puppet
- SaltStack
- PowerShell
- InSpec

Examples of Provisioners

- upload files
 - single file
 - multiple files (directory)
- execute script
 - execute single script
 - run inline commands
- ansible playbook
- powershell script

Useful Features of Provisioners

- run on specific builds
- build-specific overrides
- pause
- retries
- timeouts